

Analysis Report

Your Name

September 19, 2025

1 Introduction

This report presents the analysis of various strategies using EPLE results and visual representations through TikZ diagrams.

2 TikZ Figures

Figure 1: Addition Diagram

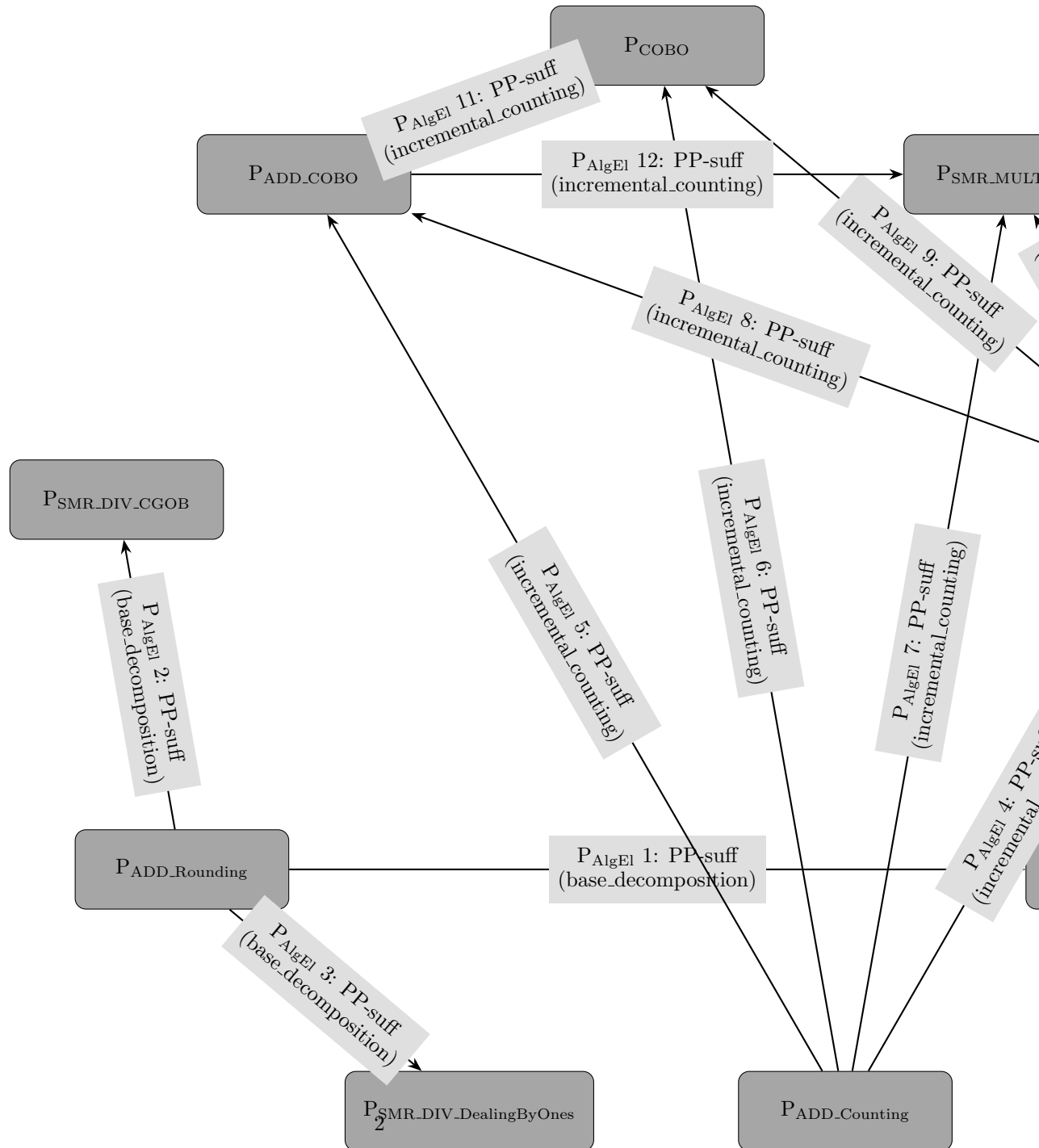
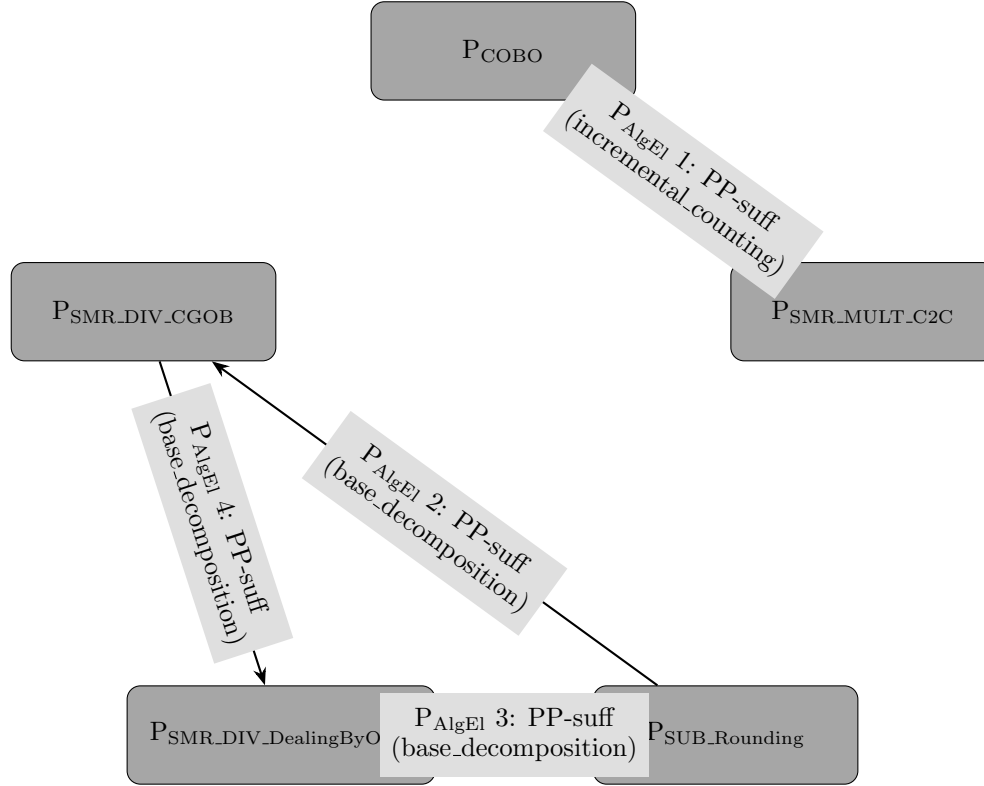


Figure 2: Miscellaneous Diagram



3 EPLE Results

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5         "type": "decomposition",
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8           "SMR_DIV_DealingByOnes",
9           "SMR_DIV_CGOB",
10          "SUB_Rounding"
11        ],
12        "usage_count": 4
13      },
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16        "strategies_using": [
17          "COBO",
18          "SMR_MULT_C2C",

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```

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20         "ADD_Counting",
21         "ADD_COBO"
22     ],
23     "usage_count": 5
24 }
25 },
26 "elaborations": [
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32         ],
33         "type": "inter_categorial",
34         "confidence": 1.0
35     },
36     {
37         "base_strategy": "ADD_Rounding",
38         "elaborated_strategy": "SMR_DIV_CGOB",
39         "shared_patterns": [
40             "base_decomposition"
41         ],
42         "type": "inter_categorial",
43         "confidence": 1.0
44     },
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50         ],
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56         "elaborated_strategy": "ADD_Chunking",
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59         ],
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63     {
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68         ],
69         "type": "intra_categorial",
70         "confidence": 1.0
71     },
72     {
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74         "elaborated_strategy": "COBO",
75         "shared_patterns": [

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77     ],
78     "type": "inter_categorical",
79     "confidence": 1.0
80 },
81 {
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83     "elaborated_strategy": "SMR_MULT_C2C",
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86     ],
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88     "confidence": 1.0
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113     ],
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115     "confidence": 1.0
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169     "confidence": 1.0
170 }
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174         "base_decomposition"
175     ],
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177     "ADD_Counting": [
178         "incremental_counting"
179     ],
180     "ADD_Chunking": [
181         "incremental_counting"
182     ],
183     "ADD_RMB": [],
184     "ADD_COBO": [
185         "incremental_counting"
186     ],
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188         "incremental_counting"
189     ],

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192     "ChunkingC": [],
193     "ChunkingB": [],
194     "ChunkingA": [],
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204     "SMR_DIV_CGOB": [
205         "base_decomposition"
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210     ],
211     "SMR_DIV_UCR": []
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217         "strategies": [
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219             "SMR_MULT_C2C",
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222             "ADD_Counting",
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224             "ADD_Rounding",
225             "SMR_DIV_CGOB",
226             "ADD_CBO"
227         ],
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pnode/.style={rectangle, rounded corners=5pt, draw, fill=gray
!70, text=black, minimum height=1.3cm, minimum width=3.5cm,
align=center, inner xsep=0.3cm, inner ysep=0.2cm},\n graybox/.
style={rectangle, fill=lightgray!50, inner sep=4pt, minimum
height=1.1cm, anchor=center, align=center, text centered},\n %
Arrow Styles\n solidarrow/.style={-Stealth, thick},\n
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textarrow/.style={align=center, inner sep=1pt}\n]\n\\tikzset{
font=\\linespread{0.8}\\selectfont}\n\n% Diagram for: addition\n
\n\n\\node[pnode] (P_0) at (0.00,9.00) {P\\textsubscript{CBO
}};\n\n\\node[pnode] (P_1) at (5.79,6.89) {P\\textsubscript{
SMR_MULT_C2C}};\n\n\\node[pnode] (P_2) at (8.86,1.56) {P\\
textsubscript{ADD_Chunking}};\n\n\\node[pnode] (P_3) at
(7.79,-4.50) {P\\textsubscript{SUB_Rounding}};\n\n\\node[pnode] (
P_4) at (3.08,-8.46) {P\\textsubscript{ADD_Counting}};\n\n\\node[
pnode] (P_5) at (-3.08,-8.46) {P\\textsubscript{

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229 SMR_DIV_DealingByOnes}};\n\node[pnode] (P_6) at (-7.79,-4.50)
230 {P\textsubscript{ADD_Rounding}};\n\node[pnode] (P_7) at
231 (-8.86,1.56) {P\textsubscript{SMR_DIV_CGOB}};\n\node[pnode] (
232 P_8) at (-5.79,6.89) {P\textsubscript{ADD_COB0}};\n\n\draw[
233 solidarrow] (P_6) -- node[graybox, midway, sloped] {P\
234 textsubscript{AlgEl} 1: PP-suff \\\ (base\_decomposition)} (
235 P_3);\n\draw[solidarrow] (P_6) -- node[graybox, midway, sloped
236 ] {P\textsubscript{AlgEl} 2: PP-suff \\\ (base\_
237 _decomposition)} (P_7);\n\draw[solidarrow] (P_6) -- node[
238 graybox, midway, sloped] {P\textsubscript{AlgEl} 3: PP-suff
239 \\\ (base\_decomposition)} (P_5);\n\draw[solidarrow] (P_4)
240 -- node[graybox, midway, sloped] {P\textsubscript{AlgEl} 4: PP
- suff \\\ (incremental\_counting)} (P_2);\n\draw[solidarrow]
(P_4) -- node[graybox, midway, sloped] {P\textsubscript{AlgEl}
} 5: PP-suff \\\ (incremental\_counting)} (P_8);\n\draw[
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(P_0);\n\draw[solidarrow] (P_4) -- node[graybox, midway,
sloped] {P\textsubscript{AlgEl} 7: PP-suff \\\ (incremental\_
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suff \\\ (incremental\_counting)} (P_0);\n\draw[solidarrow]
(P_2) -- node[graybox, midway, sloped] {P\textsubscript{AlgEl}
10: PP-suff \\\ (incremental\_counting)} (P_1);\n\draw[
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(P_0);\n\draw[solidarrow] (P_8) -- node[graybox, midway,
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pnode/.style={rectangle, rounded corners=5pt, draw, fill=gray
!70, text=black, minimum height=1.3cm, minimum width=3.5cm,
align=center, inner xsep=0.3cm, inner ysep=0.2cm},\n graybox/.
style={rectangle, fill=lightgray!50, inner sep=4pt, minimum
height=1.1cm, anchor=center, align=center, text centered},\n %
Arrow Styles\n solidarrow/.style={-Stealth, thick},\n
dashedarrow/.style={dashed, -Stealth, thick, gray},\n
textarrow/.style={align=center, inner sep=1pt}\n]\n\begin{tikzset{
font=\linespread{0.8}\selectfont}\n\n% Diagram for:
miscellaneous\n\n\node[pnode] (P_0) at (0.00,5.00) {P\
textsubscript{COB0}};\n\node[pnode] (P_1) at (4.76,1.55) {P\
textsubscript{SMR_MULT_C2C}};\n\node[pnode] (P_2) at

```



```

(2.94,-4.05) {P\\textsubscript{SUB_Rounding}};\n\\node[pnode] (
P_3) at (-2.94,-4.05) {P\\textsubscript{SMR_DIV_DealingByOnes
}};\n\\node[pnode] (P_4) at (-4.76,1.55) {P\\textsubscript{
SMR_DIV_CGOB}};\n\\n\\draw[solidarrow] (P_0) -- node[graybox,
midway, sloped] {P\\textsubscript{AlgEl} 1: PP-suff \\\ (
incremental\\_counting)} (P_1);\n\\draw[solidarrow] (P_2) --
node[graybox, midway, sloped] {P\\textsubscript{AlgEl} 2: PP-
suff \\\ (base\\_decomposition)} (P_4);\n\\draw[solidarrow] (
P_2) -- node[graybox, midway, sloped] {P\\textsubscript{AlgEl}
3: PP-suff \\\ (base\\_decomposition)} (P_3);\n\\draw[
solidarrow] (P_4) -- node[graybox, midway, sloped] {P\\
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4 Analysis Results

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9         "SUB_Rounding"
10      ],
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12    },
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19        "ADD_Counting",
20        "ADD_COBO"
21      ],
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75         "incremental_counting"
76     ],
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78     "confidence": 1.0
79 },
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83     "shared_patterns": [
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87     "confidence": 1.0
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93         "incremental_counting"
94     ],

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146     "shared_patterns": [
147         "base_decomposition"
148     ],
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150     "confidence": 1.0
151 },

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154   "elaborated_strategy": "SMR_DIV_DealingByOnes",
155   "shared_patterns": [
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157   ],
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159   "confidence": 1.0
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163   "elaborated_strategy": "SMR_DIV_DealingByOnes",
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168   "confidence": 1.0
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177     "incremental_counting"
178   ],
179   "ADD_Chunking": [
180     "incremental_counting"
181   ],
182   "ADD_RMB": [],
183   "ADD_COBO": [
184     "incremental_counting"
185   ],
186   "COBO": [
187     "incremental_counting"
188   ],
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190   "SUB_Decomposition": [],
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202   "SMR_MULT_CBO": [],
203   "SMR_DIV_CGOB": [
204     "base_decomposition"
205   ],
206   "SMR_DIV_IDR": [],
207   "SMR_DIV_DealingByOnes": [
208     "base_decomposition"

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209     ],
210     "SMR_DIV_UCR": []
211 }
212 }
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5 Conclusion

This report summarizes the findings from the analysis and visualizations provided.